Question 6: Non-Functional Requirements

Assumptions

- ±48M readings per day
- Financial billing system with 7-year retention
- Time-based pricing alignment is critical
- South African operating environment (load shedding, late data arrival)

1. Application and Data Resilience

Requirement	Target	Rationale
Availability	99.9%	Readings arrive continuously; billing runs nightly
RTO (Recovery Time Objective)	15 minutes	Prolonged downtime impacts revenue processing
RPO (Recovery Point Objective)	<10 seconds	Loss of readings results in unbilled consumption
Data durability	11 nines	Financial audit and regulatory retention

The system must tolerate:

- Data centre failure
- Database corruption
- Late arrival of readings after outages
- Infrastructure instability due to load shedding

The solution must allow **replay of readings** and **point-in-time recovery**.

2. Security (Application, Data, Access)

Security must be enforced at four layers.

Layer	Requirement

Network	Encrypted communication for all meter and user traffic
Application	Strong authentication (meters and users), role-based authorization
Data	Encryption at rest, protection of PII, masking in non-prod
Access	Least privilege, MFA for privileged roles, full access logging

Separation of duties must exist between:

- Developers
- DB administrators
- Supervisors approving readings
- Billing system consuming validated data

3. Auditability and Traceability

Because this is a financially auditable system, the following must be traceable for 7 years:

- Every reading received
- Every validation decision
- Every manual correction or approval
- Every tariff price change
- Every billing calculation

The system must support answering questions such as:

> "Why was this customer charged this amount on this date?"

This requires:

- Immutable historical records
- Version history for corrected readings
- Logging of who changed what and why
- Correlation between reading, tariff version, and invoice

4. Load and Performance Testing (Top 2 Metrics)

Metric 1: Reading Ingestion Throughput

- Sustain 1,500 readings/sec (3× average peak)
- p95 response time <200ms
- 0% data loss under sustained load

Why: Dropped readings = direct revenue loss.

Metric 2: Billing Run Completion

- Process ±50M readings in <4 hours
- 100% billing accuracy
- No locking or contention affecting throughput

Why: Billing must complete before the next business day to maintain cash flow.

5. Additional Critical NFR: Data Quality and Integrity

This is the most important NFR for this system.

Incorrect readings or misaligned tariffs lead to:

- Revenue loss
- Customer disputes
- Regulatory risk

The system must enforce:

Quality Dimension	Target
Accuracy	>99.9% validation pass rate
Completeness	>99% expected readings received daily
Consistency	Readings must align with customer and tariff history at that time
Uniqueness	No duplicate readings per meter per timestamp
Timeliness	>90% readings within 24 hours

Daily reconciliation must exist between:

- Readings received
- Readings billed
- Revenue calculated